

C. User Documentation (for Non-Technical Users)

- **App Overview:** This is an interactive web tool for brain tumor segmentation. It uses an AI model (Attention U-Net) to analyze multi-modal MRI scans of the brain and automatically identify and label different parts of a tumor. The application provides both 2D and 3D visualizations of the segmentation results.
- **Purpose and Use Cases:** The tool is designed for medical students, researchers, or clinicians to:
 - Visualize and understand the different components of a brain tumor, such as the Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET).
 - Upload their own MRI scans in NIfTI format to get an automated segmentation.
 - Explore pre-loaded example cases to see how the model performs.
 - View the tumor in a 3D interactive model to better understand its structure and location.
- **Data:** The application requires four types of MRI scans in NIfTI format (`.nii`):
 1. **FLAIR** (Fluid-Attenuated Inversion Recovery)
 2. **T1** (T1-weighted)
 3. **T1ce** (T1-weighted contrast-enhanced)
 4. **T2** (T2-weighted)

You can also upload an optional **Ground Truth** segmentation file to compare the model's prediction with a known result.

- **Step-by-Step Instructions:**

1. **Load Your Data:** When you open the application, you have two choices:
 - **Option A: Upload Your Own Files:** In the "Upload Files" tab, use the five input boxes to select each of your `.nii` or `.nii.gz` files from your computer. The system needs the four main modalities (FLAIR, T1, T1ce, T2) to run a prediction.
 - **Option B: Use an Example:** Click the "Use Examples" tab. A list of pre-loaded cases will appear. Click on a case name (e.g., "1") to automatically load its data into the application.
2. **Run Prediction:** Once all four required files are loaded, the **"Run Prediction"** button will become clickable. Click it to start the AI segmentation process. A status message will appear, indicating that the model is running. This may take over a minute to complete.
3. **View 2D Results:**
 - After the prediction is finished, you can see the results in the "2D Slice Viewer" section.
 - Click through the tabs: **"Modalities"** (to see the original scans), **"Prediction"** (to see the model's output), and **"Ground Truth"** (if you loaded one).
 - Use the **slider** at the bottom of this section to scroll through the different axial slices of the brain scans.

4. Explore the 3D View:

- Scroll down to the "3D Layers" section. An interactive 3D model of the brain and the predicted tumor will be displayed.
- **Interact:** Click and drag your mouse on the model to rotate it. Use the scroll wheel to zoom in and out.
- **Toggle Layers:** Use the checkboxes on the left (`Brain` , `Whole Tumor` , `Tumor Core` , `Enhancing Tumor`) to show or hide different components of the 3D model. This helps in isolating and viewing specific tumor regions.

• Troubleshooting:

- **"Run Prediction" button is grayed out:** This means one or more of the four required MRI scans (FLAIR, T1, T1ce, T2) have not been loaded. Please ensure a file has been selected for each.
- **Prediction Fails:** If you receive an error during prediction, it may be because the uploaded files are not in the correct NIfTI format or are corrupted. Please verify your input files.
- **3D Tumor Model Doesn't Appear:** The 3D tumor meshes will only be generated and displayed *after* a prediction has been successfully completed.

D. API Documentation (for projects exposing REST or inference endpoints)

This documents the RESTful API endpoints provided by the Flask backend.

Base URL: `http://127.0.0.1:7860`

Section	Endpoint	HTTP Method	Description	Parameters / Request Format	Example Request / Response Format
Health Check	<code>/</code>	GET	Serves the main <code>index.html</code> page.	None	Response: <code>200 OK</code> with HTML content.
Examples	<code>/list_examples</code>	GET	Retrieves a list of available example case IDs.	None	Response: <code>200 OK</code> Content-Type: <code>application/json</code> <code>["1", "case1"]</code>
Examples	<code>/load_example</code>	POST	Loads an example case into a user session on the server.	Request Body: <code>application/json</code> <code>{ "id": "1", "session_id": "user-uuid-12345" }</code>	Response: <code>200 OK</code> Content-Type: <code>application/json</code> <code>{ "message": "Example loaded", ... }</code>
Upload	<code>/upload</code>	POST	Uploads a single NIfTI file for a specific modality and session.	Request Body: <code>multipart/form-data</code> - <code>file</code> : The NIfTI file (<code>.nii</code> or <code>.nii.gz</code>). - <code>modality</code> : String (e.g., 'flair', 't1ce'). - <code>session_id</code> :	Response: <code>200 OK</code> Content-Type: <code>application/json</code> <code>{ "message": "flair uploaded", ... }</code>

				String (e.g., 'user-uuid-12345').	
Inference	/predict	POST	Runs the full segmentation pipeline on the data uploaded for a session.	Request Body: application/json { "session_id": "user-uuid-12345" }	Response: 200 OK Content-Type: application/json { "message": "Prediction successful", ... }
Visualization	/get_all_slices	GET	Fetches all 2D slices for a modality as base64-encoded PNGs.	Query Parameters: - modality : String (e.g., 't1ce', 'pred'). - session_id : String.	Response: 200 OK Content-Type: application/json { "t1ce": ["data:image/png;base64:...", ...] }
Visualization	/get_3d_mesh_data	GET	Generates and returns 3D surface mesh data for rendering.	Query Parameters: - modality : String ('t1ce' for brain, 'pred' for tumor). - session_id : String.	Response: 200 OK Content-Type: application/json { "WT": { "vertices": [...], "faces": [...], ... }